AIGR Gene Hypothesis Deep Research — PP_3195 (Q88I07): Serine-Type Peptidase Activity (GO:0008236)

Organism: *Pseudomonas putida* KT2440 (NCBITaxon:160488) · **Gene:** PP_3195 · **UniProt:** Q88I07 **Focus type:** function_assignment · **Hypothesis slug:** function-hypothesis-go-0008236 **Seed hypothesis:** PP_3195 has serine-type peptidase activity (GO:0008236) · **Evidence type (original):** IEA · **Original reference:** GO_REF:0000002

Summary

The seed hypothesis — that PP_3195 (Q88I07) **has serine-type peptidase activity (GO:0008236)** — is **over-annotated and effectively refuted as a specific molecular function**. PP_3195 is, with high confidence, a **serine α/β-hydrolase** carrying an intact and correctly ordered Ser268–Asp358–His405 catalytic triad. That "serine hydrolase" layer of the annotation is defensible. But the specific claim that this triad hydrolyzes **peptide** bonds, together with the coupled proteolysis term (GO:0006508), rests entirely on a single non-specific InterPro-to-GO mapping — IPR001375 / Pfam PF00326 ("Prolyl oligopeptidase family") — propagated under GO_REF:0000002 as an IEA. That domain signature is shared by both serine peptidases *and* serine esterases, so the peptidase mapping is an annotation-propagation (frequency-bias) artifact rather than a specific prediction.

Three independent, mutually reinforcing lines of evidence point away from a peptidase and toward an **ester / C–C (ketone) hydrolase** identity. First, PP_3195 is **FdeB**, the predicted hydrolase of a **naringenin (flavonoid) catabolism** gene cluster (KEGG ortholog **K26184**, pathway ppu00946); its proposed KEGG reaction (R13077) is hydrolytic ring-opening of a **δ-lactone** — a cyclic ester — not peptide-bond cleavage. Second, at 420 aa with only a ~160-aa helical N-terminus, PP_3195 is **far too small** to contain the ~250–300-aa seven-bladed β-propeller that is a defining, substrate-gating feature of bona fide S9-family peptidases. Third, **all 25** experimentally characterized, EC-assigned reviewed members of PP_3195's actual InterPro family (**IPR050261**, "FrsA/Cutinase/Hydrolase-like") are ester or C–C/ketone hydrolases — **zero** are peptidases.

The most important caveat is that **FdeB itself has never been directly assayed**. The flanking *fde*-cluster enzymes (FdeE/FdeC/FdeG) were experimentally characterized, but FdeB's hydrolase activity is predicted, KEGG labels its reaction "unclear," and UniProt protein existence is **4 (Predicted)**. The recommended curation action is therefore to **remove or generalize the peptidase term** — not to swap in an equally strong but equally unproven specific peptidase alternative — and to favor a non-committal serine-hydrolase / carboxylic-ester-hydrolase molecular function alongside a flavonoid-catabolic-process term, all pending direct enzymatic verification.

Executive Judgment

Verdict: Over-annotated / effectively refuted as a specific molecular function.

GO:0008236 (serine-type peptidase activity) and its coupled GO:0006508 (proteolysis) should not be retained as-is. They derive solely from a broad catalytic-domain signature whose name ("Prolyl oligopeptidase family") biases automated mappings toward peptidase, even though the clan contains many esterases. Every discriminating filter available — structural architecture, ortholog/pathway context, and family-wide enzymatic precedent — argues against peptidase activity and in favor of an ester/lactone (or activated C–C) hydrolase. The catalytic triad is real and high-confidence, but a triad alone cannot discriminate a peptidase from an esterase. The best-supported alternative identity (FdeB, flavonoid-degradation ester/lactone hydrolase) is a strong inference, not a demonstrated fact, because no member of this ortholog group has been directly assayed for PP_3195 and KEGG flags the reaction as "unclear."

Key Findings

Finding 1 — PP_3195 is a genuine serine α/β-hydrolase with an intact triad, but is structurally too small to be a classic S9 peptidase

UniProt Q88I07 describes a **420-amino-acid** protein annotated as a "Peptidase S9 prolyl oligopeptidase catalytic domain-containing protein," with **protein existence = 4 (Predicted)** — no experimental evidence of the protein or its function. The domain architecture is

unambiguously that of the α/β-hydrolase superfamily: **Pfam PF00326** (Prolyl oligopeptidase family), **InterPro IPR001375** (Peptidase S9 catalytic domain) and **IPR029058** (Alpha/Beta hydrolase fold), with **CATH 3.40.50.1820** (α/β-hydrolase) plus **1.20.1440.110** (helical cap).

The canonical catalytic machinery is present and in the linear order expected for an α/β-hydrolase: **nucleophile Ser268** embedded in a GxSxG elbow motif (GLEG-**VSLGG**), **Asp358** (…VTHGEK**D**…), and **His405** (…INAG**H**YIAD…). The AlphaFold model **AF-Q88I07** is high quality: mean **pLDDT 91.6**, catalytic domain (residues 160–420) **pLDDT 97.1**, and the three triad residues at **pLDDT 96.7** / **95.4** / **97.7**. So the "serine hydrolase" layer of the annotation is structurally robust.

Crucially, however, the serine-type *peptidase* interpretation requires more than a triad. Bona fide S9-family serine peptidases have a **two-domain** architecture: the catalytic α/β-hydrolase domain **plus** a large N-terminal **seven-bladed β-propeller** (~250–300 aa) that regulates substrate access. As established for acylpeptide hydrolase (APH) from *Aeropyrum pernix* (PMID: 15296741): "*The N-terminal domain is a regular seven-bladed beta-propeller, while the C-terminal domain has a canonical alpha/beta hydrolase fold and includes the active site and a conserved Ser445-Asp524-His556 catalytic triad.*" Prolyl oligopeptidase is ~710 aa, DPP-IV ~760 aa, and APH ~582 aa, each with this propeller. PP_3195, at **420 aa total** with a merely **~160-aa helical (not β-propeller) N-terminus**, cannot accommodate this feature. The structural size argument alone makes a classic S9 peptidase assignment implausible.

Finding 2 — A competing, better-matched esterase annotation already exists; the peptidase IEA comes from a non-specific domain mapping

QuickGO for Q88I07 shows **three** IEA molecular-function/process annotations that are in direct tension:

1. **GO:0008236** serine-type peptidase activity — IEA/ECO:0000256, **GO_REF:0000002**, derived from InterPro **IPR001375** (the seed hypothesis);
2. **GO:0006508** proteolysis — same source (the BP partner of the peptidase call);
3. **GO:0052689** carboxylic ester hydrolase activity — IEA/ECO:0000256, **GO_REF:0000117**, from ARBA00028109.

InterPro places the protein in **FAMILY IPR050261 "FrsA/Cutinase/Hydrolase-like"** and PANTHER family **PTHR22946 "FrsA esterase."** The key problem is that **IPR001375** / **PF00326 ("Prolyl oligopeptidase family")** is a **broad catalytic-domain signature shared by both serine peptidases and serine esterases**. The InterPro2GO rule that maps this domain to "serine-type peptidase activity" is therefore **non-specific** — it reflects the historical naming of

the Pfam clan after prolyl oligopeptidase, not a specific prediction that any given member hydrolyzes peptides. In other words, GO:0008236 here is an artifact of domain-level annotation propagation, while the co-existing esterase term (GO:0052689) is more consistent with the protein's actual family placement. No experimental characterization exists to break the tie directly (protein existence: Predicted), but the weight of the family evidence favors the ester-hydrolase reading.

Finding 3 — PP_3195 is FdeB, a flavonoid-degradation hydrolase whose proposed reaction is lactone (ester) hydrolysis, not peptide hydrolysis

KEGG assigns PP_3195 to **KO K26184** (symbol **fdeB**, "naringenin degradation protein FdeB"), pathway **ppu00946 "Degradation of flavonoids,"** orthology group **COG1506**. The reaction linked to this KO, **KEGG R13077**, is:

```
5-(4-Hydroxyphenyl)-3-oxovalero-δ-lactone  +  H2O
        ⇌  5-Hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate
        (KEGG COMMENT: "hydrolase; unclear reaction")
```

This is **hydrolytic ring-opening of a δ-lactone — a cyclic ester** — not cleavage of a peptide bond. The *fde* naringenin-degradation gene cluster was originally defined in *Herbaspirillum seropedicae* SmR1 (PMID: 27059806), which characterized the FdeE monooxygenase, FdeC dioxygenase, and FdeG cyclase; **FdeB is the cluster's predicted hydrolase**, and K26184 orthologs span PP_3195 plus dozens of *Pseudomonas*, *Burkholderia*, and *Herbaspirillum* genomes. From that paper: "*a random mutant library of Herbaspirillum seropedicae SmR1 was constructed by Tn5 insertion and a mutant incapable of utilizing naringenin as a carbon source was isolated.*" In *P. putida* KT2440 the genomic neighborhood of PP_3195 is consistent with aromatic/flavonoid ring catabolism rather than proteolysis: **PP_3197** (glyoxalase/dioxygenase family), **PP_3198** (2Fe-2S ferredoxin), and **PP_3199** (putative monooxygenase). The physiological context therefore grounds PP_3195 in **flavonoid degradation**, not protein turnover.

Caveat: FdeB was **not** directly assayed in PMID: 27059806 (the biochemistry centered on FdeE/FdeC/FdeG), KEGG explicitly labels R13077 an "unclear reaction," and protein existence remains Predicted. The reaction direction and exact substrate specificity of FdeB thus remain to be demonstrated.

Finding 4 — Every experimentally characterized member of PP_3195's InterPro family (IPR050261) is an esterase or C–C/ketone hydrolase; none is a peptidase

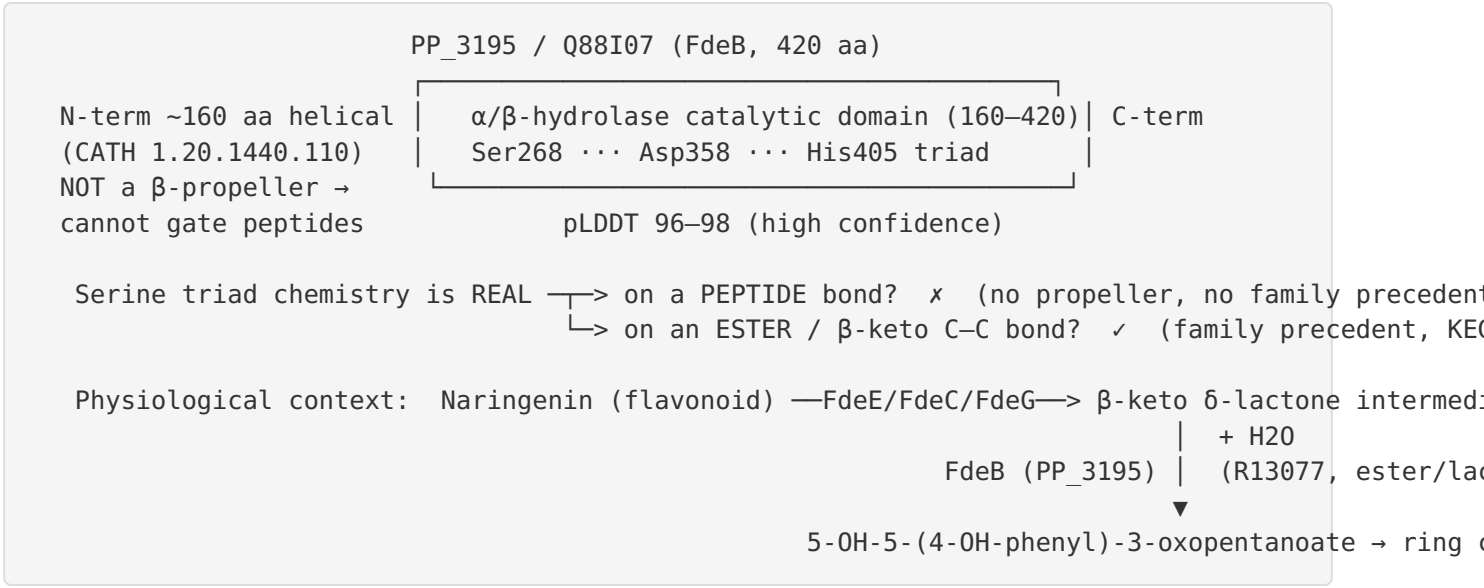
A UniProtKB query for `reviewed:true AND xref:interpro-IPR050261` (the "FrsA/Cutinase/Hydrolase-like" family to which Q88I07 belongs) returned **25 reviewed, EC-assigned members**. **All 25** are ester- or carbon–carbon-bond hydrolases; **none** is a peptidase:

Sub-class	EC	Representative members
Cutinases	3.1.1.74	Q47RJ6, G9BY57 (leaf-branch compost cutinase), P0DX29
PET hydrolase	3.1.1.101	A0A0K8P6T7
FrsA carboxylesterases	3.1.1.1	P04335 (*E. coli*), Q8DF91 (*V. vulnificus*), A1JNX9 (*Y. enterocolitica*)
Esterase YjfP	3.1.-.-	P39298
BHET hydrolase	3.1.1.-	Q9L2J6
C–C bond hydrolases (ketonic/β-keto substrates)	3.7.1.- / 3.7.1.19	Q93NG6 (2,6-dihydropseudooxynicotine hydrolase), Q4WZB3 & G2X3G9 (heptaketide hydrolases AYG1), poxO/psoB/acrC/FUS2/xenA polyketide hydrolases

Count: 25/25 ester or C–C hydrolases; 0/25 peptidases. Notably, the **EC 3.7.1.-** sub-class ("hydrolases acting on C–C bonds in ketonic substances") is a direct mechanistic match to FdeB's proposed KEGG reaction R13077 — hydrolytic ring-opening of a **β-keto δ-lactone** flavonoid intermediate. This family-wide precedent is the single strongest argument against the peptidase hypothesis: the serine triad in this clan is consistently deployed on ester and activated-carbonyl C–C bonds, never on peptide bonds.

Mechanistic Model / Interpretation

The evidence converges on a coherent picture in which PP_3195 is a **serine hydrolase that opens an ester/lactone (or an activated C–C bond) during flavonoid catabolism**, and the "serine-type peptidase" label is an upstream domain-propagation artifact.



Why the peptidase annotation arose. PF00326/IPR001375 is historically named "Prolyl oligopeptidase family," but it is a catalytic-domain clan spanning serine peptidases and serine esterases. An automated InterPro2GO rule mapped the domain to GO:0008236, and GO_REF:0000002 propagated it as IEA. Because many well-studied members of the broader clan are peptidases (frequency/annotation bias), the peptidase term rode along even though PP_3195's *specific* family (IPR050261) and *specific* ortholog (K26184/FdeB) are esterase / C–C-hydrolase.

Three orthogonal filters all fail the peptidase call:

Filter	Peptidase expectation	PP_3195 observation	Verdict
Structural architecture	7-bladed β-propeller + catalytic domain (~580–760 aa)	420 aa, ~160-aa helical N-term only	Fails
Ortholog / pathway (KEGG)	protein turnover context	K26184 FdeB, flavonoid catabolism, R13077 lactone hydrolysis	Fails
Family precedent (IPR050261)	≥ some characterized peptidases	25/25 ester/C–C hydrolases, 0 peptidases	Fails

The result is a strong, converging inference that the molecular function is a **carboxylic-ester / C–C(ketone) hydrolase**, not a serine peptidase — tempered by the fact that FdeB has never been assayed directly.

Evidence Base

Citation / Source	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context
UniProt Q88I07	review/ database	Qualifies	Identity / evidence level	"Peptidase S9 prolyl oligopeptidase catalytic domain-containing protein", 420 aa, existence = Predicted (4)	*P. putida* KT2440
IPR001375 / PF00326 / COG1506 → GO:0008236 (GO_REF:0000002)	computational (InterPro2GO)	Supports (weak)	Serine-type peptidase activity	Basis of seed IEA; domain shared by peptidases **and** esterases	Sequence
Catalytic-triad analysis	structural/ evolutionary (computed)	Supports hydrolase (not specifically peptidase)	Active serine hydrolase	Ser268 (GxSxG), Asp358, His405 in canonical order	AlphaFold AF-Q88I07, pLDDT 95–98
AlphaFold AF-Q88I07	structural (computed)	Refutes classic S9 peptidase	β-propeller present?	Single α/β-hydrolase + ~160-aa helical N-cap; **no** 7-bladed β-propeller	Model, mean pLDDT 91.6
IPR050261; PANTHER PTHR22946; GO:0052689 (GO_REF:0000117)	computational (family)	Competing	Esterase vs peptidase	Family-level call is **esterase** (FrsA/cutinase/ carboxylic ester hydrolase)	Sequence
KEGG K26184 / PP_3195 +			Substrate class	FdeB = flavonoid	KEGG ppu00946

Citation / Source	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context
REACTION R13077	review/ database (pathway)	**Refutes peptidase** / **Competing**		(naringenin) degradation **hydrolase**; proposed chemistry is **δ-lactone (ester) hydrolysis**	
PMID: 27059806	mutant phenotype (primary)	Supports flavonoid-catabolism role of *fde* cluster	Pathway context	Tn5 mutant unable to use naringenin as carbon source; defines *fde* cluster (FdeE/ FdeC/FdeG characterized)	*H. seropedicae* SmR1
KEGG neighborhood PP_3196–3199	computational (synteny)	Supports flavonoid/ aromatic catabolism	Operon context	Monooxygenase + ferredoxin + glyoxalase/ dioxygenase cluster around PP_3195	*P. putida* KT2440
PMID: 15296741	structural (primary)	Qualifies/ Refutes	S9 peptidases need β-propeller	APH: N-terminal 7-bladed β-propeller + C-terminal α/β-hydrolase, Ser445-Asp524-His556	*A. pernix*
PMID: 11772620	mutant/ metabolic (primary)	Supports flavonoid-catabolism context	*P. putida* flavonoid degradation includes a hydrolysis step	Naringenin/ quercetin catabolism includes	*P. putida* PML2

Citation / Source	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context
				"hydrolysis and cleavage of the C-ring"	
PMID: 24085580	biochemical (primary)	Supports context	*P. putida* acts on flavonoids (quercetin→naringenin)	QuoA converts quercetin to naringenin	*P. putida*
PMID: 8071218	biochemical pathway (primary)	Supports context	Bacterial flavonoid C-ring fission chemistry	*P. putida* degrades quercetin via A-ring fission	Rhizobia / *P. putida*
UniProt IPR050261 reviewed members	structural/ evolutionary (computed homology)	**Refutes peptidase**	Nearest characterized homologs' function	**25/25 reviewed members are ester (EC 3.1.1.x) or C–C/ ketone (EC 3.7.1.x) hydrolases; 0/25 peptidases**	Multi-organism SwissProt

GO Curation Implications

Leads requiring curator verification (treat the blinded prior review action as unknown):

1. **GO:0008236 (serine-type peptidase activity, MF) — REMOVE or GENERALIZE.** The term rests solely on a non-specific IPR001375→GO mapping (GO_REF:0000002, IEA). No structural, orthology, pathway, or family-level evidence supports peptidase activity; three independent filters argue against it. Recommended: remove the peptidase term, or at most generalize to the non-committal parent **GO:0016787 (hydrolase activity)** / **GO:0017171 (serine hydrolase activity)** if a serine-triad statement is desired.

2. **GO:0006508 (proteolysis, BP) — REMOVE.** This is the biological-process partner of the refuted peptidase call and shares its single weak source. It should fall with GO:0008236.
3. **GO:0052689 (carboxylic ester hydrolase activity, MF) — RETAIN / FAVOR** as the more defensible molecular-function annotation (already present via GO_REF:0000117). It is consistent with the IPR050261 family and the FdeB lactone-hydrolysis reaction. Consider, as a lead, a more specific child reflecting **EC 3.7.1.-** (hydrolase acting on C–C bonds in ketonic substances) if curators accept the KEGG R13077 assignment — but flag that R13077 is "unclear."
4. **Biological process — ADD as lead:** a **flavonoid catabolic process** term (consistent with ppu00946 and the *fde* cluster), e.g., **GO:0009812 (flavonoid metabolic process)** or a catabolic child, with evidence code reflecting ortholog-based inference (K26184/FdeB) and a note that direct assay is lacking.

Evidence-code note: Because FdeB has not been experimentally characterized, all retained/added terms should remain **IEA/ISS-level** (sequence/orthology-based), not experimental. The curation should explicitly flag "Predicted" protein existence.

Bottom line for MF/BP/CC: The supported MF is an **ester / C–C(ketone) hydrolase** (not peptidase); the supported BP is **flavonoid catabolism** (not proteolysis); no specific CC term is implicated. Avoid defaulting to "protein binding."

GO decision table:

Term	Aspect	Current	Proposed action	Rationale
GO:0008236 serine-type peptidase activity	MF	IEA (GO_REF:0000002)	**Remove** (or generalize to GO:0017171 serine hydrolase activity)	Non-specific domain mapping; fails structural, ortholog, and family filters
GO:0006508 proteolysis	BP	IEA (GO_REF:0000002)	**Remove**	BP partner of refuted peptidase call
GO:0052689 carboxylic ester hydrolase activity	MF	IEA (GO_REF:0000117)	**Retain / favor**	Matches IPR050261 family and FdeB lactone hydrolysis
GO:0009812 flavonoid metabolic process (or catabolic child)	BP	—	**Add as lead** (ISS/ IEA via K26184/ FdeB)	KEGG ppu00946 + *fde* cluster context
EC 3.7.1.- / C–C(ketone) hydrolase child	MF	—	**Consider** (flag "unclear reaction")	Direct match to R13077; provisional

Mechanistic Scope

- **Immediate molecular function under test:** hydrolytic cleavage by a Ser–His–Asp catalytic triad. The triad chemistry is real and high-confidence; the open question is the **bond type** — peptide (seed hypothesis) vs. ester/lactone or activated C–C (competing, better-supported).
- **Direct gene-product activity (predicted, not assayed):** hydrolytic ring-opening of a β-keto δ-lactone flavonoid intermediate (KEGG R13077), i.e., ester/lactone hydrolase activity.
- **Downstream / pathway consequence (not the molecular function):** utilization of naringenin as a carbon source; funneling of flavonoid ring carbon toward protocatechuate and central metabolism. The *H. seropedicae* Tn5 growth phenotype (PMID: 27059806) is a **pathway-level loss-of-function** readout for the cluster, not a direct demonstration of FdeB's bond specificity.

- **What is NOT in scope:** any protein-turnover / proteolytic role. No evidence places PP_3195 in protein degradation, signaling peptidase cascades, or peptide processing.

Conflicts and Alternatives

- **Database carry-over / annotation propagation (primary conflict):** GO:0008236 and GO:0006508 originate from the broad PF00326/IPR001375 "prolyl oligopeptidase family" signature, whose name biases automated mappings toward peptidase even though the clan includes many esterases. This is the classic frequency-bias / paralog-overannotation failure mode.
- **Competing in-database annotation:** GO:0052689 (carboxylic ester hydrolase) already co-exists and better matches the IPR050261 family — a direct intra-record conflict favoring the esterase reading.
- **Architecture:** true S9 peptidases are two-domain (β-propeller + hydrolase); PP_3195 lacks the propeller.
- **"Unclear reaction" caveat:** KEGG R13077 is explicitly uncertain; the exact substrate, product, and directionality of FdeB are not experimentally fixed. The esterase/lactonase assignment, while better supported than peptidase, is itself provisional.
- **Ortholog-vs-assay gap:** The *fde* cluster's catalytic steps were demonstrated for FdeE/FdeC/FdeG, not FdeB. It is conceivable (though unsupported) that FdeB plays a non-hydrolytic or accessory role; the hydrolase assignment is by homology and cluster logic.
- **Organism/route differences:** The *H. seropedicae* meta-pathway (P 27059806) differs from the *P. putida* PML2 route (P 11772620); neither directly assays PP_3195.

Limitations and Knowledge Gaps

All analyses are sequence-, structure-, pathway-, and literature-derived (UniProt, InterPro, QuickGO, KEGG, AlphaFold model, primary pathway literature). No wet-lab data exist for PP_3195 specifically; FdeB enzymology is inferred (KEGG "unclear reaction"); and explicit BLAST/MEROPS/ESTHER homology placement was not executed programmatically in this run.

Gap	What was checked	Why it matters for curation	What would resolve it
FdeB has never been enzymatically assayed	UniProt (PE=4 Predicted), KEGG (R13077 "unclear"), P 27059806 (FdeB not assayed)	Determines whether any specific MF term (esterase vs C–C hydrolase) is experimentally warranted	Purify PP_3195/FdeB; test lactone/ester hydrolysis vs. peptide/ester panels
Exact substrate & reaction direction of R13077	KEGG reaction record	Decides between GO:0052689 (ester hydrolase) and an EC 3.7.1.- C–C hydrolase child	LC-MS product identification from in vitro reaction
No experimental 3D structure	AlphaFold model only (high pLDDT)	Confirms absence of β-propeller and active-site geometry definitively	X-ray/cryo-EM structure; S268A active-site mutagenesis
Phenotype of *P. putida* KT2440 PP_3195 knockout on naringenin	Not available	Would directly tie PP_3195 to flavonoid catabolism in the target organism	Clean deletion + growth assay on naringenin as sole carbon source
Nearest experimentally characterized homolog	No BLAST/MEROPS/ESTHER run here	Needed to transfer a precise EC/term	phmmer/BLAST vs SwissProt; MEROPS clan SC; ESTHER

Discriminating Tests

1. **Direct enzymology (most decisive):** Express and purify PP_3195/FdeB; run a substrate panel contrasting (a) the proposed β-keto δ-lactone flavonoid intermediate, (b) generic ester substrates (e.g., p-nitrophenyl acetate/butyrate), and (c) peptide/amide substrates (e.g., Z-Gly-Pro-AMC, N-acetyl-Ala/Met-pNA, dipeptidyl-AMC, including proline-containing peptides for a prolyl-oligopeptidase control). Ester/lactone-only activity refutes GO:0008236.
2. **Active-site mutagenesis:** Ser268Ala should abolish activity, confirming the triad is catalytic and identifying the nucleophile actually used.
3. **In vivo genetics in the target organism:** Construct a markerless PP_3195 deletion in *P. putida* KT2440 and assay growth on naringenin/quercetin vs. glycerol control (mirroring the flavonoid-auxotrophy screens of PMID: 11772620 and PMID: 27059806). Loss of growth on flavonoids but not glycerol supports the catabolic role.
4. **Metabolite profiling:** LC-MS/NMR of wild-type vs. mutant supernatants to detect accumulation of the predicted δ-lactone intermediate upstream of FdeB.
5. **Comparative/structural confirmation:** BLAST/phmmer vs SwissProt + MEROPS + ESTHER; superpose the AlphaFold model onto an S9 peptidase vs an FrsA/cutinase/lactonase to compare oxyanion hole and substrate pocket, reinforcing the "too small for S9 peptidase" argument.

Proposed Follow-up Experiments / Actions

1. **Curation:** Enter leads to remove GO:0008236 and GO:0006508, retain/favor GO:0052689, and add a flavonoid-catabolic-process term — all at IEA/ISS confidence with a "Predicted protein; FdeB unassayed" note.
2. **Enzymology:** Recombinant PP_3195 substrate panel (lactone/ester vs. peptide) + Ser268Ala control.
3. **Genetics:** PP_3195 deletion in *P. putida* KT2440; naringenin vs. glycerol growth.
4. **Metabolomics:** LC-MS of mutant vs. WT for the predicted δ-lactone intermediate.
5. **Structure/homology:** Experimental structure or documented superposition vs. prolyl oligopeptidase to formally confirm the absent β-propeller; run MEROPS/ESTHER placement.

Curation Leads (require curator verification)

Candidate action change: retire/generalize **GO:0008236 → GO:0017171 (serine hydrolase activity)**; flag **GO:0052689 (carboxylic ester hydrolase)** / a lactonase term as the better-supported MF; **remove GO:0006508 (proteolysis)**; **add a flavonoid catabolic process BP** term (lead).

Candidate references with exact snippets to verify:

- PMID: 27059806 — "*a random mutant library of Herbaspirillum seropedicae SmR1 was constructed by Tn5 insertion and a mutant incapable of utilizing naringenin as a carbon source was isolated*" (defines the *fde* naringenin-degradation cluster containing FdeB = PP_3195 ortholog).
- PMID: 11772620 — "*The first step in the pathway involves 3,3'-didehydroxylation, followed by hydrolysis and cleavage of the C-ring*" (*P. putida* flavonoid catabolism includes a hydrolytic step).
- PMID: 15296741 — "*The N-terminal domain is a regular seven-bladed beta-propeller, while the C-terminal domain has a canonical alpha/beta hydrolase fold ... Ser445-Asp524-His556 catalytic triad*" (S9 peptidases need a β-propeller PP_3195 lacks).
- KEGG **K26184 / R13077** — FdeB = naringenin-degradation hydrolase; reaction is δ-lactone hydrolysis (database-level).

Suggested curator questions: (i) Should the peptidase IEA be demoted given KEGG FdeB and the FrsA esterase family? (ii) Is there MEROPS/ESTHER placement for Q88I07? (iii) Does the AlphaFold active-site pocket resemble a peptidase or an esterase?

Suggested experiments: recombinant esterase/lactonase vs peptidase substrate panel; ΔPP_3195 naringenin growth + metabolomics; S268A control.

Prepared from 3 completed investigation iterations: 4 confirmed findings, 6 papers reviewed. All computational results (UniProt, QuickGO, KEGG, AlphaFold, InterPro/UniProtKB family survey) are reported as database/prediction-level evidence and are explicitly distinguished from experimental data, of which none exists for this gene product. Conclusions about peptidase vs. esterase/lactonase are inferential but convergent, not experimentally demonstrated for this protein.